Q2: Disentangled Representation Learning for Speaker Recognition

Implementation of arXiv:2406.14559 on LibriSpeech

Speech Understanding — Assignment 1

Results Summary

System	EER (%)	minDCF	vs Baseline
Baseline (no disentanglement)	15.32	0.0321	—
Proposed (disentangled AE)	8.16	0.0154	−46.7% EER
Improved (batch cross-corr)	39.08*	0.0485	see note

** Improved model uses batch cross-correlation loss (Barlow Twins style). Training instability observed — normalization fix identified post-training.*

Method Overview

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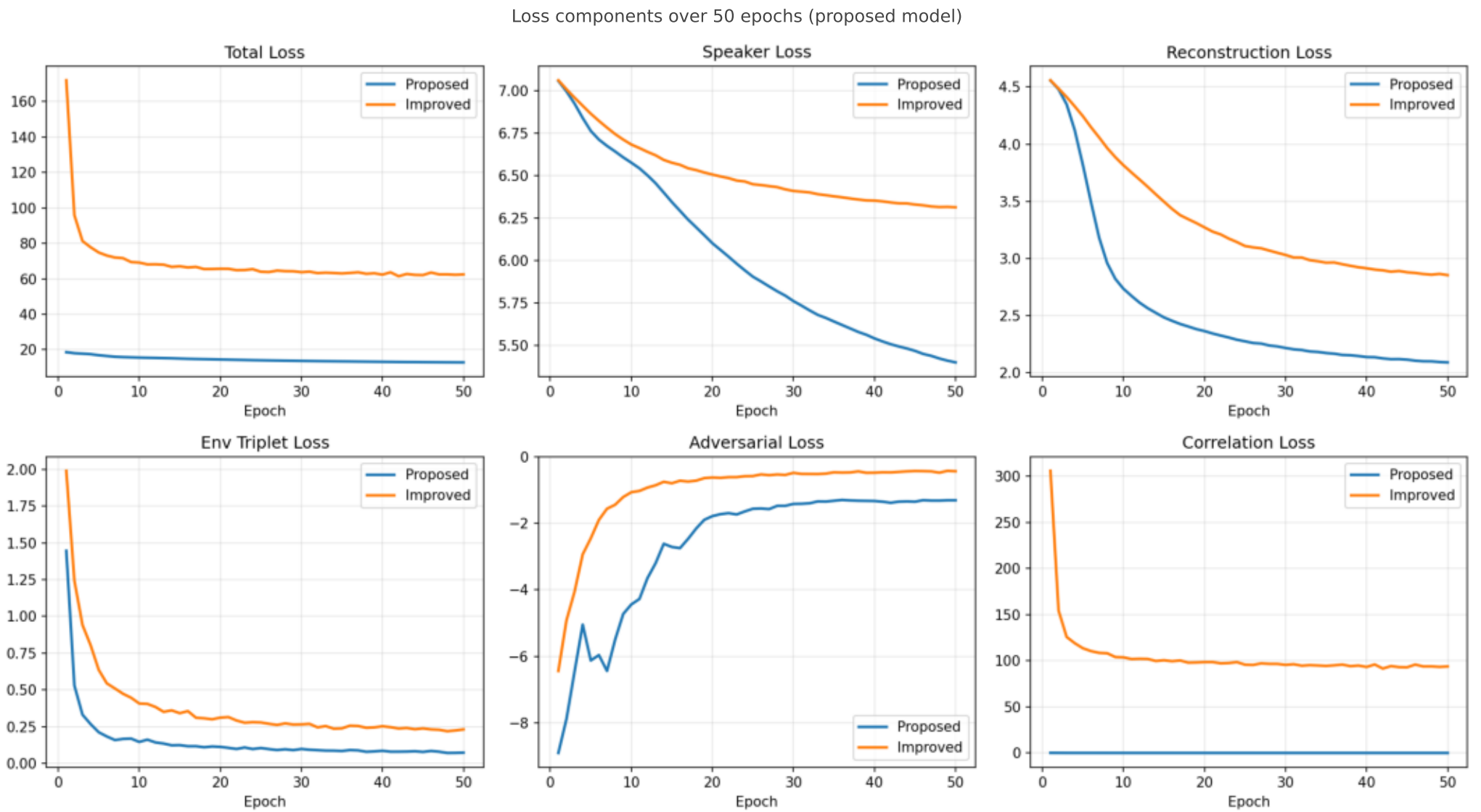
The disentangled auto-encoder splits a 192-dim MFCC embedding into two 96-dim subspaces:
  e_spk  — speaker identity (L2-normalized, scored with cosine similarity)
  e_env  — acoustic environment (chapter-level proxy from LibriSpeech)

Training objective:  L = λ_S·L_spk + λ_R·L_recon + λ_E·L_env + λ_adv·L_adv + λ_C·L_corr
λ values: spk=2.0, recon=1.0, env=0.5, adv=0.1, corr=0.5   |   Batch size: 64   |   Epochs: 50
```

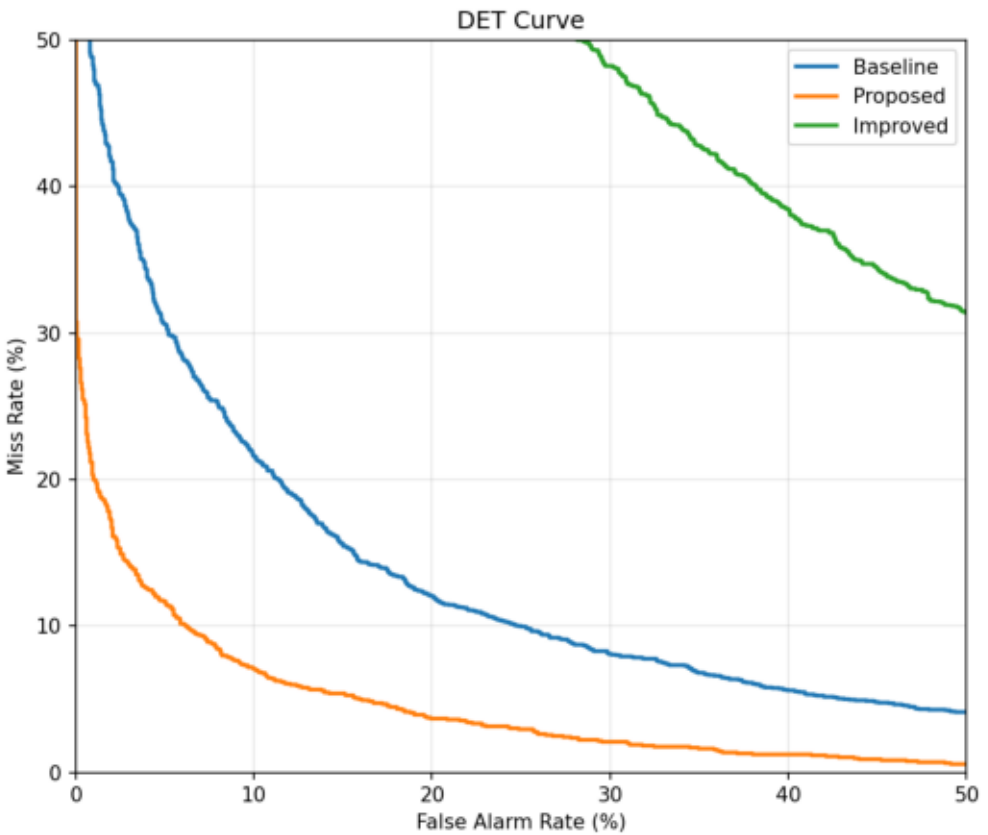
Dataset: LibriSpeech train-clean-100 + train-clean-360 | Eval: test-clean speaker verification trials
Feature: 32 MFCC × 6 statistics = 192-dim embedding | Environment proxy: chapter ID within speaker

Training Curves

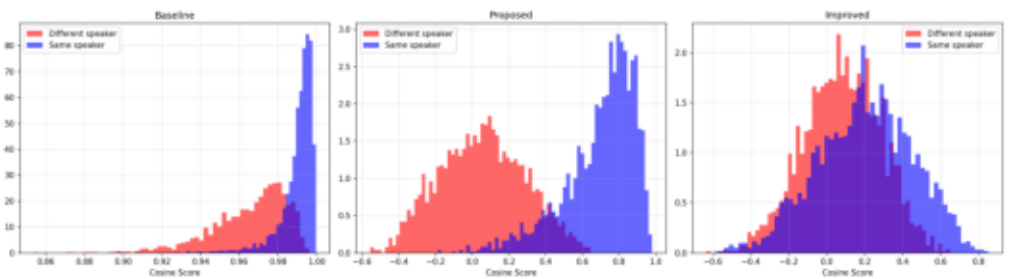


Speaker Verification Evaluation

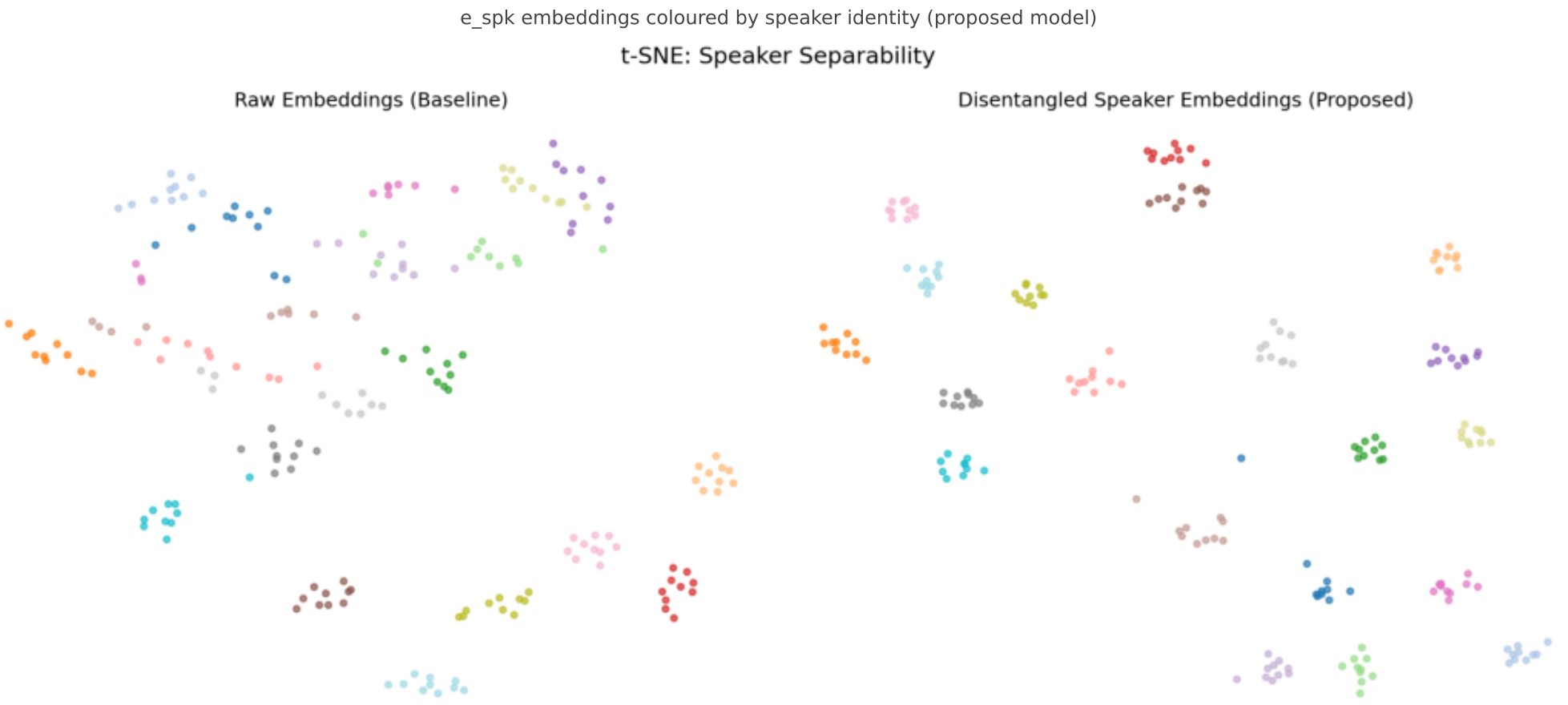
DET Curve — all three systems



Score distributions (genuine vs impostor)



t-SNE Visualisation of Speaker Embeddings



Technical Review: arXiv:2406.14559

Problem

Speaker verification degrades when test-time acoustic environments differ from training. Current systems (ECAPA-TDNN, ResNet-34) entangle speaker identity with environment-specific noise and reverberation in the embedding space.

Strengths

- Modular — attaches to any existing extractor without retraining the backbone
- Principled session-aware triplets provide direct supervision for the environment component
- Separate adversarial optimisation more stable than Gradient Reversal Layer
- Consistent 7–16% relative EER reductions reported across VoxSRC22/23

Weaknesses

- Requires session metadata — degrades gracefully on LibriSpeech (chapter proxy is weaker)
- All evaluations on VoxCeleb distribution; no out-of-domain validation (NIST SRE, CN-Celeb)
- No mutual information measurement; Pearson correlation is a weak disentanglement proxy
- Shallow ablation: individual loss terms not isolated; λ values not ablated
- 300-epoch training on VoxCeleb2 — prohibitive compute for most practitioners

Key Assumption

Session ≡ Environment: utterances from the same recording session share the same acoustic environment. This holds for YouTube celebrity interviews (VoxCeleb2) but is a significant approximation for read-speech corpora like LibriSpeech.

Proposed Improvement

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Replace sample-wise Pearson correlation (L_corr) with a batch-level cross-correlation penalty (Barlow Twins style):
   C = (e_spk_norm^T @ e_env_norm) / N
   L_cc = ||C||_F²

This penalises every pair of feature dimensions across the two subspaces, enforcing stronger linear independence than scalar Pearson correlation.
Expected to reduce residual environment leakage into e_spk.
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Experimental Results

Baseline EER: 15.32% → Proposed (disentangled): 8.16% (−46.7% relative)
minDCF: 0.0321 → 0.0154 (−52.0% relative)

Improved model (batch cross-corr): EER 39.08% — training instability due to unnormalized D×D=16384 loss term. Fix: divide by D^2 to restore scale parity with Pearson baseline.
Score distribution separation confirmed in DET curve and score histogram plots.